

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Logarithmic Power Integral Equation

Consider a function f of an arbitrary variable t . The function $f(t)$ is defined by the following property, involving an integral:

$$f(t) = \int_1^{\infty} \frac{\ln(x)}{x^t} dx, \quad t \in \mathbb{R}_{\geq 2}$$

Now consider an arbitrary constant α , where $\alpha \in \mathbb{R}_{\geq 2}$. There holds a property involving $f(t)$ and α :

$$\int_{\alpha}^{\infty} f(t) dt = 0.6\overline{7}$$

Find the exact value of α .

Solution: We begin by deriving an explicit closed-form expression for the function $f(t)$ for $t \geq 2$:

$$f(t) = \int_1^{\infty} x^{-t} \ln(x) dx$$

We apply integration by parts. Let:

$$u = \ln(x) \implies du = \frac{1}{x} dx$$

$$dv = x^{-t} dx \implies v = \frac{x^{1-t}}{1-t} = -\frac{1}{t-1} x^{-(t-1)}$$

Using the integration by parts formula $\int u dv = uv - \int v du$:

$$f(t) = \left[-\frac{\ln(x)}{(t-1)x^{t-1}} \right]_1^{\infty} + \frac{1}{t-1} \int_1^{\infty} x^{-t} dx$$

Since $t \geq 2$, the upper boundary term vanishes as $x \rightarrow \infty$ because polynomial decay dominates logarithmic growth ($\lim_{x \rightarrow \infty} \frac{\ln(x)}{x^{t-1}} = 0$). At the lower limit $x = 1$, $\ln(1) = 0$. Thus, the boundary terms evaluate to zero:

$$f(t) = 0 + \frac{1}{t-1} \left[-\frac{1}{t-1} x^{-(t-1)} \right]_1^{\infty} = \frac{1}{(t-1)^2} (0 - (-1)) = \frac{1}{(t-1)^2}$$

Now we substitute this expression into the second integral equation involving the parameter α :

$$\int_{\alpha}^{\infty} f(t) dt = \int_{\alpha}^{\infty} \frac{1}{(t-1)^2} dt$$

Evaluating this improper integral directly:

$$\int_{\alpha}^{\infty} (t-1)^{-2} dt = \left[-\frac{1}{t-1} \right]_{\alpha}^{\infty} = 0 - \left(-\frac{1}{\alpha-1} \right) = \frac{1}{\alpha-1}$$

We are given that this integral equals the recurring decimal $0.\overline{67}$. We convert this decimal to a rational fraction:

$$S = 0.\overline{67} = 0.676767\ldots \implies 100S = 67.6767\ldots \implies 99S = 67 \implies S = \frac{67}{99}$$

Equating our integrated result to this fraction:

$$\frac{1}{\alpha - 1} = \frac{67}{99} \implies \alpha - 1 = \frac{99}{67}$$

Solving algebraically for α :

$$\alpha = 1 + \frac{99}{67} = \frac{166}{67}$$

Since 67 is a prime number, the fraction cannot be simplified further. The exact value is:

$$\alpha = \frac{166}{67}$$

1.2 Hard Integral

Logarithmic Rational Sextic Integral

Evaluate the definite integral:

$$\int_0^{\infty} \frac{\ln(x)}{x^6 - 1} dx$$

Solution: Let the integral over the positive real line be denoted by I :

$$I = \int_0^{\infty} \frac{\ln(x)}{x^6 - 1} dx$$

We split the interval of integration at $x = 1$:

$$I = \int_0^1 \frac{\ln(x)}{x^6 - 1} dx + \int_1^{\infty} \frac{\ln(x)}{x^6 - 1} dx$$

For the second integral over $(1, \infty)$, we perform the inversion substitution $x = \frac{1}{u}$, which gives $dx = -\frac{1}{u^2} du$. As $x \rightarrow 1$, $u \rightarrow 1$, and as $x \rightarrow \infty$, $u \rightarrow 0$:

$$\int_1^{\infty} \frac{\ln(x)}{x^6 - 1} dx = \int_1^0 \frac{\ln(1/u)}{(1/u)^6 - 1} \left(-\frac{1}{u^2}\right) du = \int_0^1 \frac{-\ln(u)}{\frac{1-u^6}{u^6}} \left(\frac{1}{u^2}\right) du = \int_0^1 \frac{-\ln(u)u^4}{1-u^6} du$$

For the first integral over $(0, 1)$, we rewrite the denominator as $x^6 - 1 = -(1 - x^6)$:

$$\int_0^1 \frac{\ln(x)}{x^6 - 1} dx = \int_0^1 \frac{-\ln(x)}{1 - x^6} dx$$

Combining both integrals under a common domain $(0, 1)$ with dummy variable x :

$$I = \int_0^1 \frac{-\ln(x)(1 + x^4)}{1 - x^6} dx$$

We can evaluate this elegantly by utilizing the standard parametric integral identity valid for $0 < a < b$:

$$\int_0^\infty \frac{x^{a-1}}{1-x^b} dx = \frac{\pi}{b} \csc\left(\frac{a\pi}{b}\right) \quad \left(\text{or } \frac{\pi}{b} \cot\left(\frac{a\pi}{b}\right) \text{ depending on sign convention}\right)$$

More directly, differentiating the well-known Mellin transform identity:

$$\int_0^\infty \frac{x^{a-1}}{x^b-1} dx = -\frac{\pi}{b} \cot\left(\frac{a\pi}{b}\right)$$

with respect to the parameter a yields:

$$\frac{\partial}{\partial a} \int_0^\infty \frac{x^{a-1}}{x^b-1} dx = \int_0^\infty \frac{x^{a-1} \ln(x)}{x^b-1} dx = -\frac{\pi}{b} \frac{\partial}{\partial a} \left[\cot\left(\frac{a\pi}{b}\right) \right]$$

Using $\frac{d}{dz} \cot(z) = -\csc^2(z)$:

$$\int_0^\infty \frac{x^{a-1} \ln(x)}{x^b-1} dx = \left(\frac{\pi}{b}\right)^2 \csc^2\left(\frac{a\pi}{b}\right)$$

For our specific integral I , we have $b = 6$ and the numerator factor is $\ln(x) = x^{1-1} \ln(x)$, corresponding precisely to $a = 1$. Since the parameter condition $0 < 1 < 6$ is strictly satisfied, we substitute $a = 1$ and $b = 6$:

$$I = \left(\frac{\pi}{6}\right)^2 \csc^2\left(\frac{\pi}{6}\right)$$

We evaluate the trigonometric term:

$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2} \implies \csc\left(\frac{\pi}{6}\right) = 2 \implies \csc^2\left(\frac{\pi}{6}\right) = 4$$

Substituting this back into the expression for I :

$$I = \frac{\pi^2}{36} \times 4 = \frac{\pi^2}{9}$$

The exact closed-form value of the integral is:

$$I = \frac{\pi^2}{9}$$

Differentials

2.1 Medium Differential

First-Order Linear Trigonometric Equation

Solve the differential equation:

$$y' + y \tan(x) = \sin(x)$$

Given $y(0) = 2$, find $y\left(\frac{\pi}{4}\right)$.

Solution: This is a first-order linear differential equation of the standard form $y' + P(x)y = Q(x)$, where $P(x) = \tan(x)$ and $Q(x) = \sin(x)$. We compute the integrating factor $I(x)$:

$$I(x) = e^{\int P(x) dx} = e^{\int \tan(x) dx} = e^{-\ln|\cos(x)|} = \sec(x)$$

Multiplying both sides of the differential equation by the integrating factor $\sec(x)$:

$$\sec(x)y' + \sec(x)\tan(x)y = \sec(x)\sin(x)$$

The left-hand side condenses into the derivative of a product, while the right-hand side simplifies via $\sec(x)\sin(x) = \tan(x)$:

$$\frac{d}{dx} [y \sec(x)] = \tan(x)$$

Integrating both sides with respect to x :

$$y \sec(x) = \int \tan(x) dx = \ln|\sec(x)| + C$$

Multiplying through by $\cos(x)$ yields the general solution:

$$y(x) = \cos(x) \ln|\sec(x)| + C \cos(x)$$

We apply the initial condition $y(0) = 2$ to determine the constant of integration C :

$$y(0) = \cos(0) \ln|\sec(0)| + C \cos(0) \implies 2 = 1 \cdot \ln(1) + C(1) \implies 2 = 0 + C \implies C = 2$$

Thus, the specific solution satisfying the initial condition is:

$$y(x) = \cos(x) \ln(\sec(x)) + 2 \cos(x)$$

We are required to evaluate this function at $x = \frac{\pi}{4}$:

$$\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}, \quad \sec\left(\frac{\pi}{4}\right) = \sqrt{2}$$

Substituting these trigonometric values into our solution:

$$y\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} \ln(\sqrt{2}) + 2 \left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{2} \ln(2^{1/2}) + \sqrt{2} = \frac{\sqrt{2}}{4} \ln(2) + \sqrt{2}$$

Factoring out $\sqrt{2}$, the exact value is:

$$y\left(\frac{\pi}{4}\right) = \sqrt{2} \left(1 + \frac{1}{4} \ln(2)\right)$$

2.2 Hard Differential

Non-Linear Homogeneous Polar Differential Equation

Solve the differential equation:

$$\frac{x dx + y dy}{y dx - x dy} = x^2 + 2y^2 + \frac{y^4}{x^2}$$

Express the solution curve in terms of polar coordinates $r(\theta)$, where $r \geq 0$ is the radial distance and θ is the angle. Given $r(0) = \frac{1}{\sqrt{2}}$, find the exact value of $r\left(\frac{\pi}{4}\right)$.

Solution: We transform the differential equation into polar coordinates using the standard relations $x = r \cos \theta$ and $y = r \sin \theta$, where $r^2 = x^2 + y^2$ and $\tan \theta = \frac{y}{x}$. Differentiating these relations yields the fundamental polar differential identities:

$$x dx + y dy = \frac{1}{2} d(x^2 + y^2) = \frac{1}{2} d(r^2) = r dr$$

$$x dy - y dx = r^2 d\theta \implies y dx - x dy = -r^2 d\theta$$

Substituting these expressions into the left-hand side of our differential equation:

$$\frac{x dx + y dy}{y dx - x dy} = \frac{r dr}{-r^2 d\theta} = -\frac{1}{r} \frac{dr}{d\theta}$$

Next, we algebraically simplify the right-hand side using a common denominator:

$$x^2 + 2y^2 + \frac{y^4}{x^2} = \frac{x^4 + 2x^2y^2 + y^4}{x^2} = \frac{(x^2 + y^2)^2}{x^2}$$

Converting this rational expression into polar coordinates:

$$\frac{(r^2)^2}{(r \cos \theta)^2} = \frac{r^4}{r^2 \cos^2 \theta} = r^2 \sec^2 \theta$$

Equating the simplified left- and right-hand sides produces a separable differential equation in r and θ :

$$-\frac{1}{r} \frac{dr}{d\theta} = r^2 \sec^2 \theta \implies \frac{dr}{d\theta} = -r^3 \sec^2 \theta$$

We separate variables to isolate r and θ :

$$\frac{dr}{r^3} = -\sec^2 \theta d\theta$$

Integrating both sides:

$$\int r^{-3} dr = -\int \sec^2 \theta d\theta \implies -\frac{1}{2r^2} = -\tan \theta + C \implies \frac{1}{2r^2} = \tan \theta - C$$

We apply the initial condition $r(0) = \frac{1}{\sqrt{2}}$ at $\theta = 0$:

$$\frac{1}{2\left(1/\sqrt{2}\right)^2} = \tan(0) - C \implies \frac{1}{2(1/2)} = 0 - C \implies 1 = -C \implies C = -1$$

Substituting $C = -1$ back into the integrated equation gives the polar curve:

$$\frac{1}{2r^2} = \tan \theta + 1 \implies r^2(\theta) = \frac{1}{2(\tan \theta + 1)}$$

Since radial distance is non-negative ($r \geq 0$), we take the positive principal square root:

$$r(\theta) = \frac{1}{\sqrt{2(\tan \theta + 1)}}$$

To find the required value $r\left(\frac{\pi}{4}\right)$, we evaluate this function at $\theta = \frac{\pi}{4}$, noting that $\tan\left(\frac{\pi}{4}\right) = 1$:

$$r\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2(1+1)}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

The exact value is:

$$r\left(\frac{\pi}{4}\right) = \frac{1}{2}$$

Derivatives

3.1 Medium Derivative

Inverse Function Derivative of Inverse Trigonometric Sum

Let $(f^{-1})'$ denote the derivative of the inverse function. If $f(x) = \tan^{-1}(x) + \sin^{-1}\left(\frac{x}{2}\right)$, compute $(f^{-1})'\left(\frac{2\pi}{3}\right)$.

Solution: By the standard derivative theorem for inverse functions, the derivative of $f^{-1}(y)$ at a point y_0 is given by the reciprocal of the derivative of $f(x)$ at the corresponding point x_0 :

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)} \quad \text{where } f(x_0) = y_0$$

****Step 1:** Determine the point x_0 such that $f(x_0) = \frac{2\pi}{3}$ ****** We set up the equation:

$$\tan^{-1}(x_0) + \sin^{-1}\left(\frac{x_0}{2}\right) = \frac{2\pi}{3}$$

Because both $\tan^{-1}(x)$ and $\sin^{-1}\left(\frac{x}{2}\right)$ are strictly increasing functions over the domain $[-2, 2]$, their sum $f(x)$ is strictly increasing and therefore bijective. There exists a unique real root x_0 . Testing the standard algebraic value $x_0 = \sqrt{3}$:

$$\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$$

Summing these angles:

$$f(\sqrt{3}) = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$$

Thus, the unique corresponding input is $x_0 = \sqrt{3}$.

****Step 2:** Compute the first derivative $f'(x)$ ****** We differentiate $f(x)$ using standard inverse trigonometric differentiation formulas and the chain rule:

$$f'(x) = \frac{d}{dx} \left[\tan^{-1}(x) \right] + \frac{d}{dx} \left[\sin^{-1}\left(\frac{x}{2}\right) \right] = \frac{1}{1+x^2} + \frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}} \cdot \frac{1}{2} = \frac{1}{1+x^2} + \frac{1}{\sqrt{4-x^2}}$$

****Step 3:** Evaluate $f'(x_0)$ and find the inverse derivative ****** We evaluate the derivative at $x_0 = \sqrt{3}$:

$$f'(\sqrt{3}) = \frac{1}{1+(\sqrt{3})^2} + \frac{1}{\sqrt{4-(\sqrt{3})^2}} = \frac{1}{1+3} + \frac{1}{\sqrt{4-3}} = \frac{1}{4} + \frac{1}{1} = \frac{5}{4}$$

Taking the reciprocal yields the required inverse derivative:

$$(f^{-1})'\left(\frac{2\pi}{3}\right) = \frac{1}{f'(\sqrt{3})} = \frac{1}{5/4} = \frac{4}{5}$$

3.2 Hard Derivative

Fifth Derivative of Logarithmic Power Product

Let $f(x) = x \ln^5(x)$. Find $f^{(5)}(e)$.

Solution: We compute the fifth derivative $f^{(5)}(x)$ evaluated at $x = e$ using the General Leibniz Rule for the n th derivative of a product of two functions:

$$(uv)^{(n)} = \sum_{k=0}^n \binom{n}{k} u^{(n-k)} v^{(k)}$$

Let $u(x) = \ln^5(x)$ and $v(x) = x$. Notice that since $v(x)$ is a linear function of x , its higher-order derivatives vanish identically:

$$v(x) = x, \quad v'(x) = 1, \quad v''(x) = v'''(x) = v^{(4)}(x) = v^{(5)}(x) = 0$$

Therefore, the Leibniz summation collapses to just the first two non-zero terms ($k = 0$ and $k = 1$):

$$f^{(5)}(x) = \binom{5}{0} u^{(5)}(x) v(x) + \binom{5}{1} u^{(4)}(x) v'(x) = x u^{(5)}(x) + 5 u^{(4)}(x)$$

We now systematically compute the successive derivatives of $u(x) = \ln^5(x)$ using the chain rule and product rule:

- **First derivative $u'(x)$:**

$$u'(x) = 5 \ln^4(x) \cdot \frac{1}{x} = 5x^{-1} \ln^4(x)$$

- **Second derivative $u''(x)$:**

$$u''(x) = -5x^{-2} \ln^4(x) + 5x^{-1} \left(4 \ln^3(x) \cdot \frac{1}{x} \right) = x^{-2} \left[-5 \ln^4(x) + 20 \ln^3(x) \right]$$

- **Third derivative $u'''(x)$:**

$$\begin{aligned} u'''(x) &= -2x^{-3} \left[-5 \ln^4(x) + 20 \ln^3(x) \right] + x^{-2} \left[-\frac{20 \ln^3(x)}{x} + \frac{60 \ln^2(x)}{x} \right] \\ &= x^{-3} \left[10 \ln^4(x) - 60 \ln^3(x) + 60 \ln^2(x) \right] \end{aligned}$$

- **Fourth derivative $u^{(4)}(x)$:**

$$\begin{aligned} u^{(4)}(x) &= -3x^{-4} \left[10 \ln^4(x) - 60 \ln^3(x) + 60 \ln^2(x) \right] + x^{-3} \left[\frac{40 \ln^3(x)}{x} - \frac{180 \ln^2(x)}{x} + \frac{120 \ln(x)}{x} \right] \\ &= x^{-4} \left[-30 \ln^4(x) + 220 \ln^3(x) - 360 \ln^2(x) + 120 \ln(x) \right] \end{aligned}$$

- **Fifth derivative $u^{(5)}(x)$:**

$$\begin{aligned} u^{(5)}(x) &= -4x^{-5} \left[-30 \ln^4(x) + 220 \ln^3(x) - 360 \ln^2(x) + 120 \ln(x) \right] \\ &\quad + x^{-4} \left[-\frac{120 \ln^3(x)}{x} + \frac{660 \ln^2(x)}{x} - \frac{720 \ln(x)}{x} + \frac{120}{x} \right] \\ &= x^{-5} \left[120 \ln^4(x) - 1000 \ln^3(x) + 2100 \ln^2(x) - 1200 \ln(x) + 120 \right] \end{aligned}$$

Now we evaluate these higher derivatives specifically at $x = e$, utilizing the identity $\ln(e) = 1$:

$$u^{(4)}(e) = e^{-4} \left[-30(1)^4 + 220(1)^3 - 360(1)^2 + 120(1) \right] = \frac{-30 + 220 - 360 + 120}{e^4} = -\frac{50}{e^4}$$

$$u^{(5)}(e) = e^{-5} \left[120(1)^4 - 1000(1)^3 + 2100(1)^2 - 1200(1) + 120 \right] = \frac{120 - 1000 + 2100 - 1200 + 120}{e^5} =$$

Finally, we substitute $u^{(4)}(e)$ and $u^{(5)}(e)$ back into our Leibniz formula for $f^{(5)}(e)$:

$$f^{(5)}(e) = e \cdot u^{(5)}(e) + 5 \cdot u^{(4)}(e) = e \left(\frac{140}{e^5} \right) + 5 \left(-\frac{50}{e^4} \right) = \frac{140}{e^4} - \frac{250}{e^4} = -\frac{110}{e^4}$$

The exact closed-form value is:

$$f^{(5)}(e) = -\frac{110}{e^4}$$

Physics & Engineering

4.1 Mechanics

Supervillain Ray Gun Radiation Pressure Power

You are a supervillain constructing a ray gun, having decided that conventional bullets are far too pedestrian. A normal bullet exerts a force of approximately $F = 5 \times 10^4 \text{ N}$ and a human is an almost perfectly absorbing target. Take the speed of light $c = 3.00 \times 10^8 \text{ m s}^{-1}$. Determine the minimum power your ray gun must have in order to produce an effect comparable to that of an ordinary firearm, expressed in Terawatts (TW).

Solution: When a collimated electromagnetic beam (such as a laser or ray gun) strikes a perfectly absorbing target, photons transfer all of their linear momentum to the target upon absorption. For a single photon of energy E , its relativistic momentum is:

$$p = \frac{E}{c}$$

For a total electromagnetic energy U delivered over a time duration t , the total momentum transferred to the target is:

$$P_{\text{mom}} = \frac{U}{c}$$

By Newton's Second Law of Motion, the force F exerted by the radiation pressure on the target equals the time rate of change of linear momentum:

$$F = \frac{dP_{\text{mom}}}{dt} = \frac{1}{c} \frac{dU}{dt}$$

Since the beam power P_{beam} is defined as the rate of energy delivery $\frac{dU}{dt}$, the force is directly proportional to beam power:

$$F = \frac{P_{\text{beam}}}{c}$$

To achieve a mechanical effect comparable to an ordinary firearm, the ray gun must exert a force of $F = 5 \times 10^4 \text{ N}$. Solving for the required minimum power P_{beam} :

$$P_{\text{beam}} = F \cdot c = (5 \times 10^4 \text{ N}) \times (3.00 \times 10^8 \text{ m s}^{-1})$$

Performing the numerical multiplication:

$$P_{\text{beam}} = 15 \times 10^{12} \text{ Watts} = 1.5 \times 10^{13} \text{ W}$$

We express the result in Terawatts, where $1 \text{ TW} = 10^{12} \text{ W}$:

$$P_{\text{beam}} = \frac{15 \times 10^{12} \text{ W}}{10^{12} \text{ W/TW}} = 15 \text{ TW}$$

The minimum required power of the ray gun is exactly:

$$P_{\text{beam}} = 15 \text{ TW}$$

4.2 Quantum Mechanics

Neutral Pion Decay Photon Wavelength Kinematics

A neutral pion (π^0) at rest decays into two photons. The pion mass is $m = 134.98 \text{ MeV}/c^2$. Calculate the wavelength λ of each photon. Express the result as a multiple of 10^{-14} m . Use $h = 6.626 \times 10^{-34} \text{ J s} = 3.00 \times 10^8 \text{ m s}^{-1}$ and $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$.

Solution: We analyze the decay process $\pi^0 \rightarrow \gamma + \gamma$ using relativistic energy and momentum conservation. Because the neutral pion is initially at rest, its initial linear momentum is zero. By conservation of linear momentum, the two emitted photons must travel in diametrically opposite directions with equal momentum magnitudes:

$$\vec{p}_{\gamma 1} + \vec{p}_{\gamma 2} = 0 \implies |\vec{p}_{\gamma 1}| = |\vec{p}_{\gamma 2}|$$

Since for a massless photon $E_\gamma = |\vec{p}_\gamma|c$, both photons must carry equal energy ($E_{\gamma 1} = E_{\gamma 2} = E_\gamma$). By conservation of total relativistic energy:

$$E_{\text{initial}} = E_{\text{final}} \implies mc^2 = 2E_\gamma \implies E_\gamma = \frac{1}{2}mc^2$$

Given the rest energy of the pion $mc^2 = 134.98 \text{ MeV}$, the energy of each individual photon is:

$$E_\gamma = \frac{134.98 \text{ MeV}}{2} = 67.49 \text{ MeV}$$

We convert this energy from Mega-electronvolts into standard SI units of Joules ($1 \text{ MeV} = 10^6 \text{ eV}$):

$$E_\gamma = (67.49 \times 10^6 \text{ eV}) \times (1.602 \times 10^{-19} \text{ J/eV})$$

$$E_\gamma = 108.11898 \times 10^{-13} \text{ J} = 1.0811898 \times 10^{-11} \text{ J}$$

By the Planck-Einstein wave-particle relation, the wavelength λ of a photon is related to its energy by:

$$E_\gamma = \frac{hc}{\lambda} \implies \lambda = \frac{hc}{E_\gamma}$$

We compute the numerator constant hc :

$$hc = (6.626 \times 10^{-34} \text{ J s}) \times (3.00 \times 10^8 \text{ m s}^{-1}) = 19.878 \times 10^{-26} \text{ J m} = 1.9878 \times 10^{-25} \text{ J m}$$

Substituting our values into the wavelength formula:

$$\lambda = \frac{19.878 \times 10^{-26} \text{ J m}}{108.11898 \times 10^{-13} \text{ J}} = \left(\frac{19.878}{108.11898} \right) \times 10^{-13} \text{ m}$$

Performing the division:

$$\frac{19.878}{108.11898} \approx 0.1838529$$

$$\lambda \approx 0.1838529 \times 10^{-13} \text{ m} = 1.838529 \times 10^{-14} \text{ m}$$

Rounding to three significant figures, the wavelength of each photon expressed as a multiple of 10^{-14} m is:

$$\lambda \approx 1.84 \times 10^{-14} \text{ m}$$

4.3 Electromagnetism

Electrostatic Hoop Bead Small Oscillation Frequency

Two beads, each with charge q and mass m , are on a horizontal, frictionless, non-conducting, circular hoop of radius R . One of the beads is glued to the hoop at some point, while the other one performs small oscillations about its equilibrium position along the hoop. The square of the angular frequency of the small oscillations is given by $\frac{q^2}{n\pi\epsilon_0 R^3 m}$, where ϵ_0 is the permittivity of free space. Find the value of n .

Solution: Let the circular hoop lie in the horizontal xy -plane with its center at the origin and radius R . Let the glued bead (Bead 1) be fixed at angular position $\theta_1 = 0$, with Cartesian coordinates $(R, 0)$. Let the movable bead (Bead 2) be at angular position θ along the hoop, with coordinates $(R \cos \theta, R \sin \theta)$.

Because both beads possess identical charge q , they experience mutual electrostatic repulsion. The stable equilibrium position for Bead 2 occurs at the diametrically opposite point on the hoop, corresponding to angle $\theta_0 = \pi$. Let Bead 2 be displaced by a small angular deviation ϕ from equilibrium, such that:

$$\theta = \pi + \phi$$

The straight-line chord distance r between the two beads is determined by the geometry of the circle:

$$r = \sqrt{(R \cos \theta - R)^2 + (R \sin \theta)^2} = \sqrt{2R^2(1 - \cos \theta)} = 2R \sin \left(\frac{\theta}{2} \right)$$

By Coulomb's Law, the electrostatic repulsive force F_c acting along the straight line connecting the two charges is:

$$F_c = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2} = \frac{q^2}{16\pi\epsilon_0 R^2 \sin^2 \left(\frac{\theta}{2} \right)}$$

Because the hoop is rigid and frictionless, the normal force cancels any radial component of the electrostatic force. Only the tangential component $F_{\tan}$ along the circular hoop drives the motion of Bead 2. In an isosceles triangle formed by the hoop's center and the two beads, the base angles are $\frac{\pi - \theta}{2}$. The tangent line to the hoop at Bead 2 is perpendicular to the radius vector. Therefore, the angle between the chord line of action and the tangential direction along the hoop is $\alpha = \frac{\theta}{2}$. The restoring tangential force acting in the direction of decreasing displacement ϕ is:

$$F_{\tan} = -F_c \cos \left(\frac{\theta}{2} \right) = -\frac{q^2 \cos \left(\frac{\theta}{2} \right)}{16\pi\epsilon_0 R^2 \sin^2 \left(\frac{\theta}{2} \right)}$$

We now apply small-angle approximations about the equilibrium position $\theta = \pi + \phi$:

$$\frac{\theta}{2} = \frac{\pi}{2} + \frac{\phi}{2}$$

$$\sin\left(\frac{\theta}{2}\right) = \sin\left(\frac{\pi}{2} + \frac{\phi}{2}\right) = \cos\left(\frac{\phi}{2}\right) \approx 1$$

$$\cos\left(\frac{\theta}{2}\right) = \cos\left(\frac{\pi}{2} + \frac{\phi}{2}\right) = -\sin\left(\frac{\phi}{2}\right) \approx -\frac{\phi}{2}$$

Substituting these approximations into our expression for tangential force:

$$F_{\text{tan}} \approx -\frac{q^2\left(-\frac{\phi}{2}\right)}{16\pi\epsilon_0 R^2(1)^2} = -\frac{q^2}{32\pi\epsilon_0 R^2}\phi$$

We formulate Newton's Second Law of Motion along the one-dimensional circular arc $s = R\phi$:

$$ma_{\text{tan}} = F_{\text{tan}} \implies mR\frac{d^2\phi}{dt^2} = -\frac{q^2}{32\pi\epsilon_0 R^2}\phi$$

Dividing through by mR yields the differential equation for simple harmonic motion:

$$\frac{d^2\phi}{dt^2} + \left(\frac{q^2}{32\pi\epsilon_0 R^3 m}\right)\phi = 0$$

Comparing this characteristic equation $\frac{d^2\phi}{dt^2} + \omega^2\phi = 0$ with the given formula for the square of the angular frequency:

$$\omega^2 = \frac{q^2}{32\pi\epsilon_0 R^3 m} \equiv \frac{q^2}{n\pi\epsilon_0 R^3 m}$$

We equate the denominators to determine the integer constant n :

$$n = 32$$